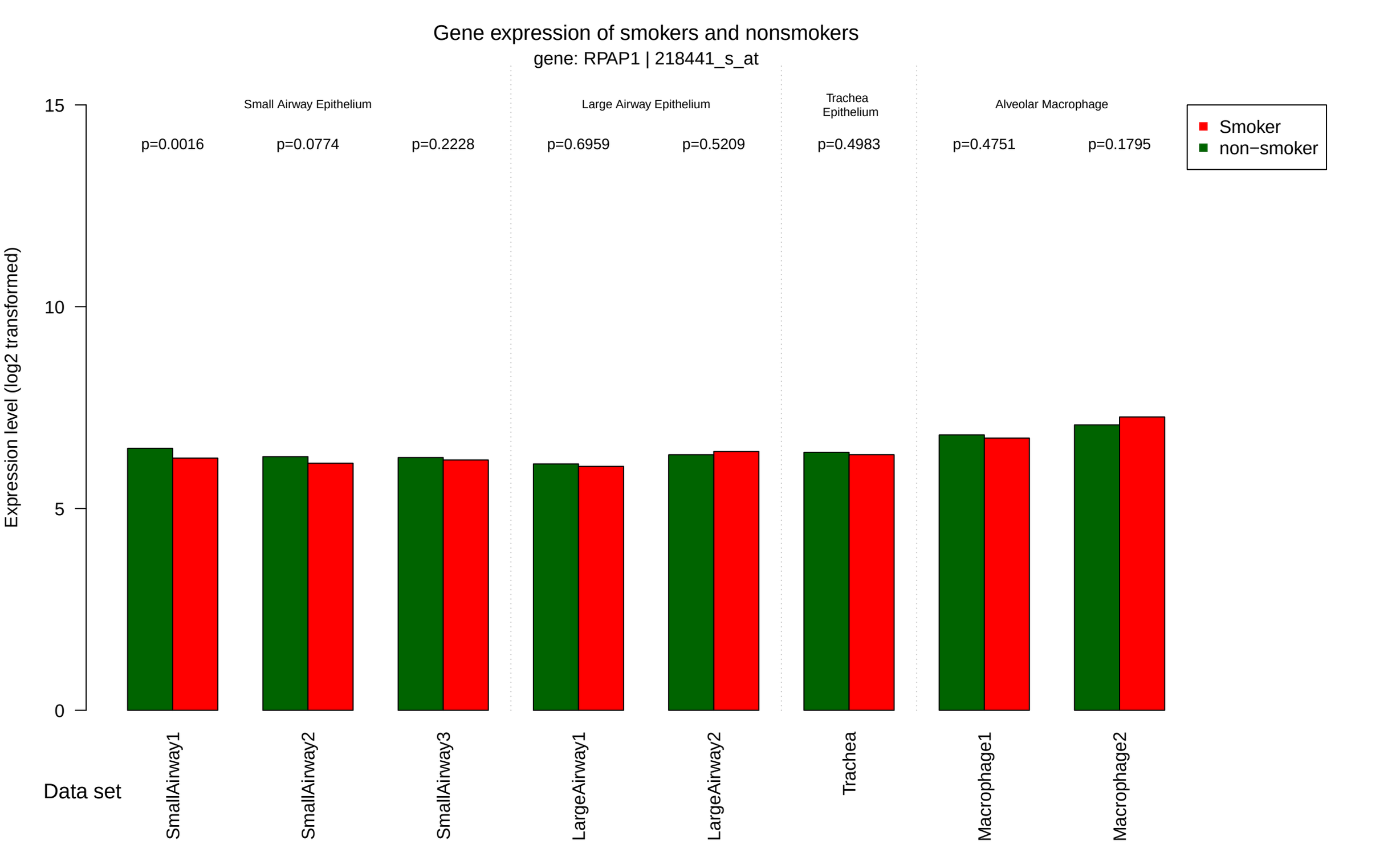
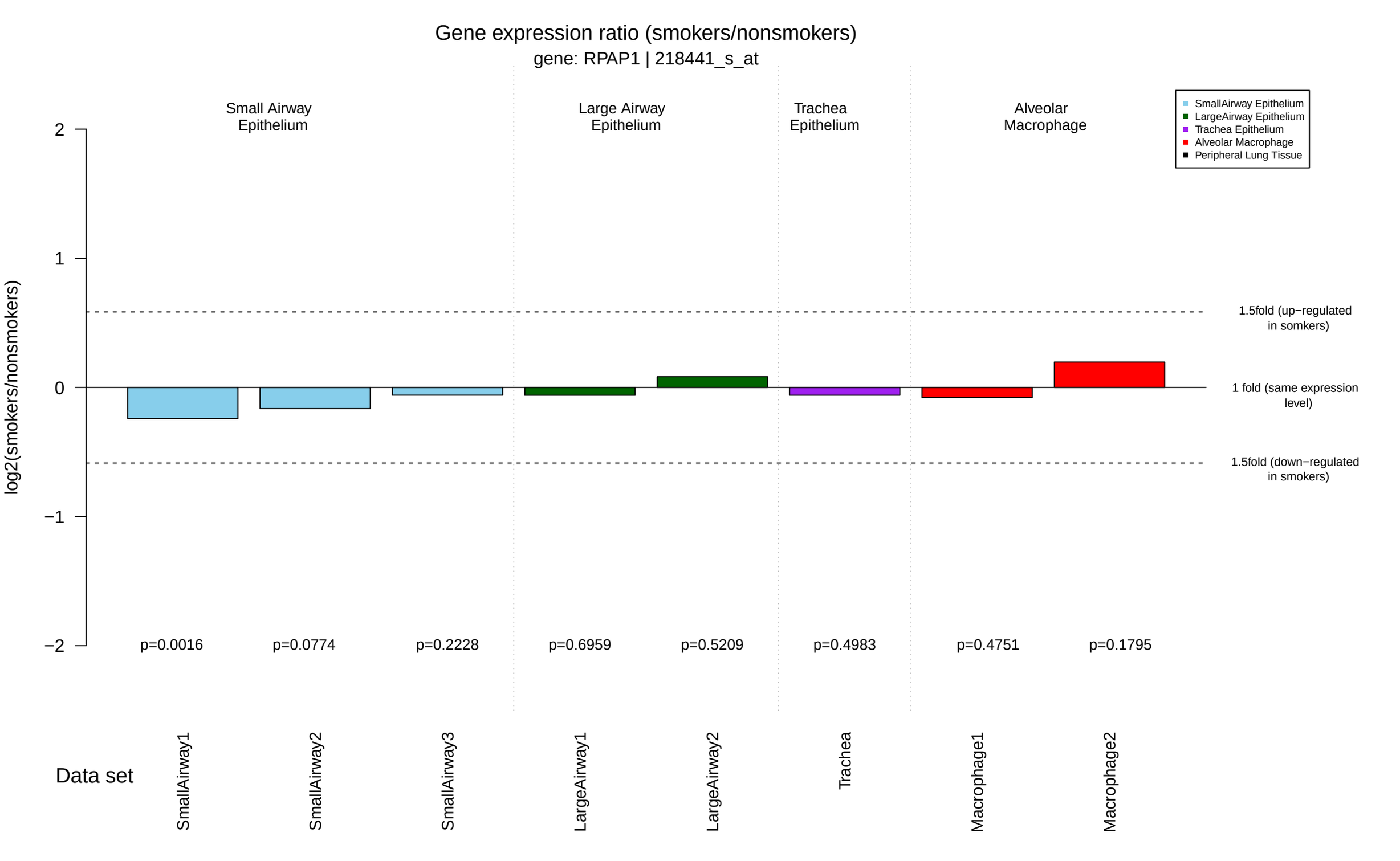
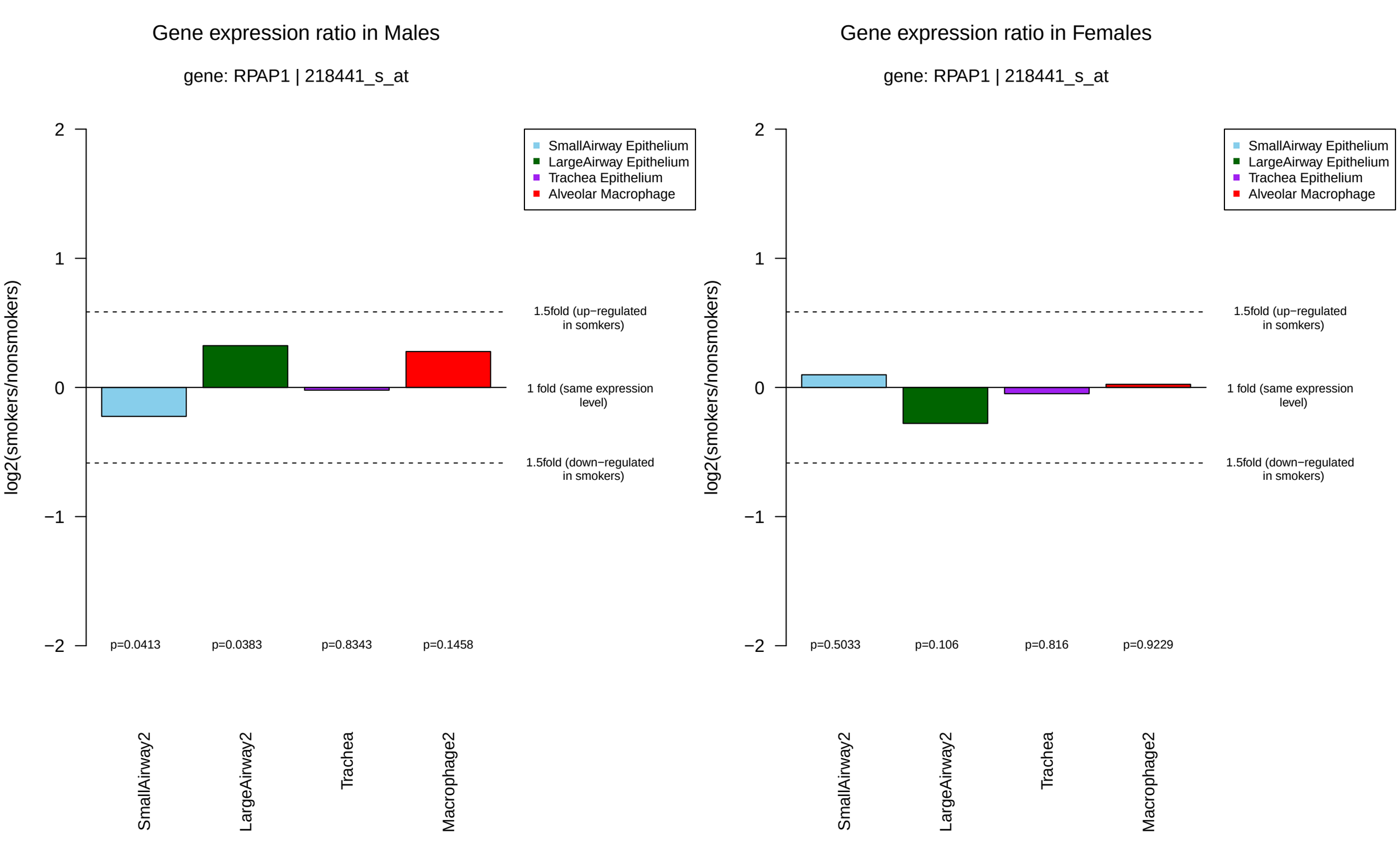
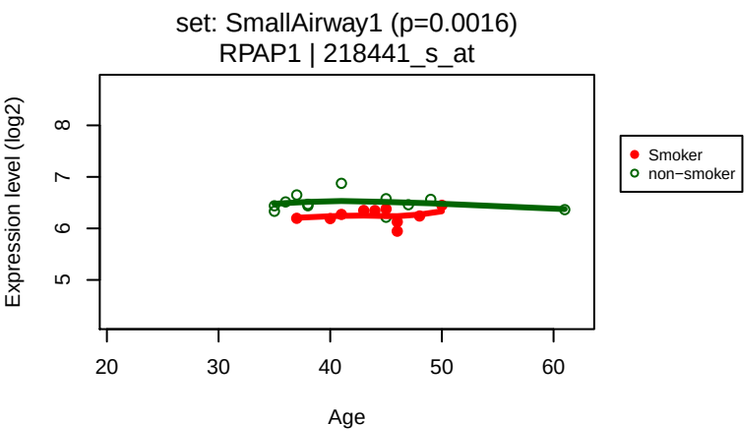
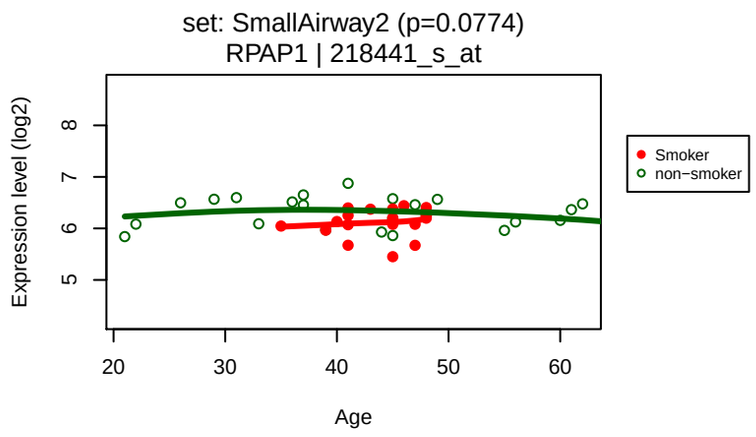
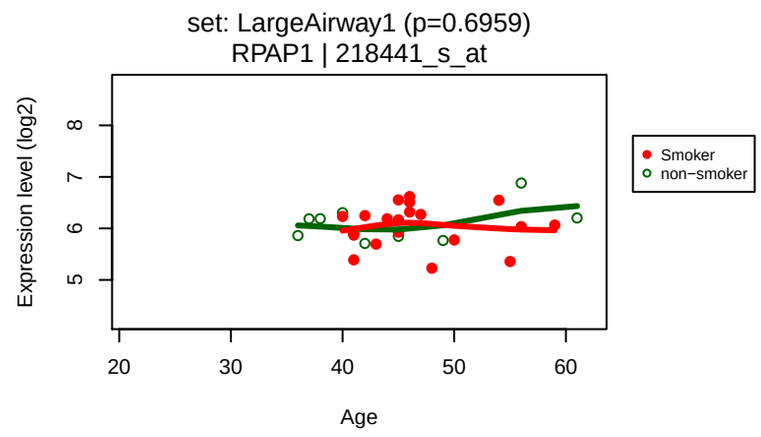

Gene expression of smokers and nonsmokers
gene: RPAP1 | 218441_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0016
p=0.0774
p=0.2228
p=0.6959
p=0.5209
p=0.4983
p=0.4751
p=0.1795
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

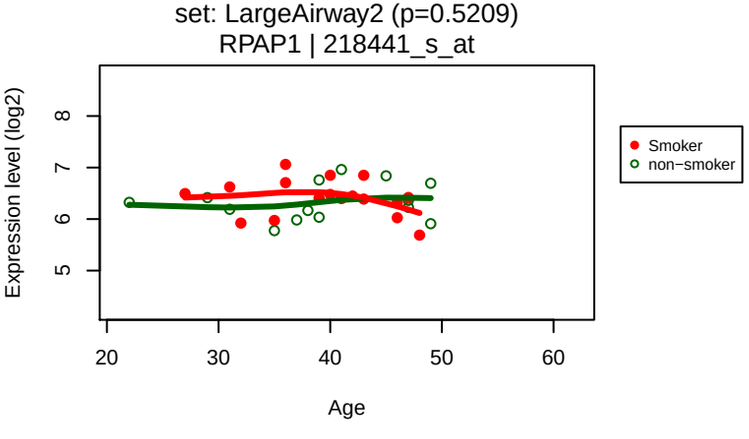

Gene expression ratio (smokers/nonsmokers)
gene: RPAP1 | 218441_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0016
p=0.0774
p=0.2228
p=0.6959
p=0.5209
p=0.4983
p=0.4751
p=0.1795
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

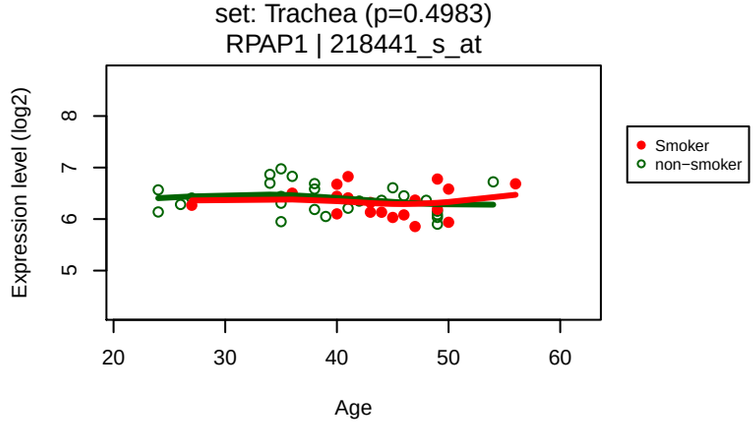

Gene expression ratio in Males
gene: RPAP1 | 218441_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0413
p=0.0383
p=0.8343
p=0.1458
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPAP1 | 218441_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5033
p=0.106
p=0.816
p=0.9229
SmallAirway2
LargeAirway2
Trachea
Macrophage2

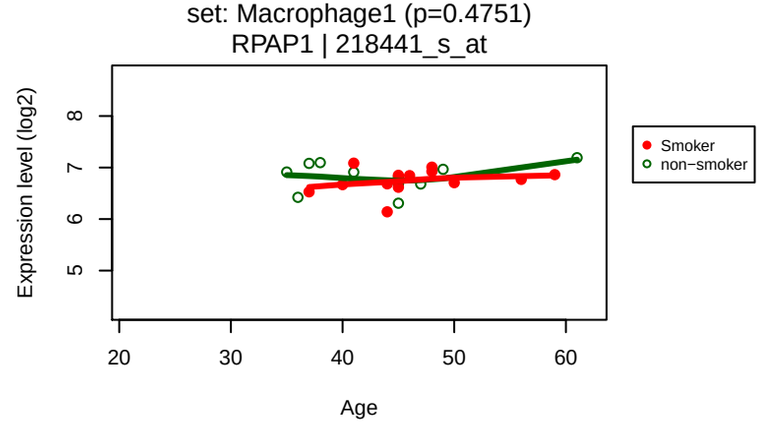

set: SmallAirway1 (p=0.0016)
RPAP1 | 218441_s_at
Expression level (log2)
Age
Smoker
non−smoker

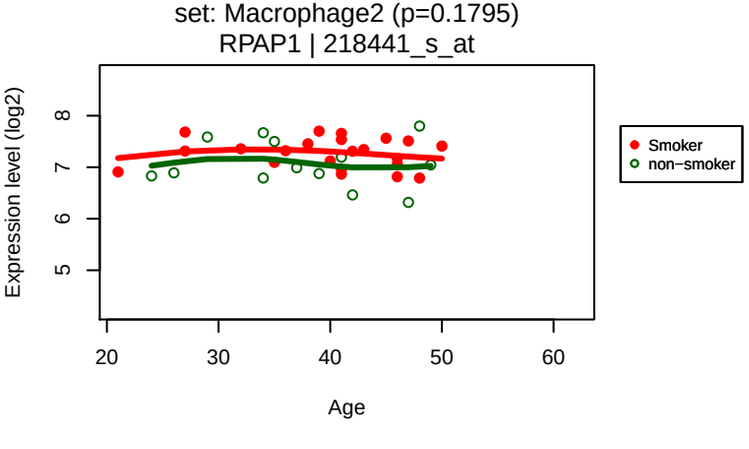

set: SmallAirway2 (p=0.0774)
RPAP1 | 218441_s_at
Expression level (log2)
Age
Smoker
non−smoker

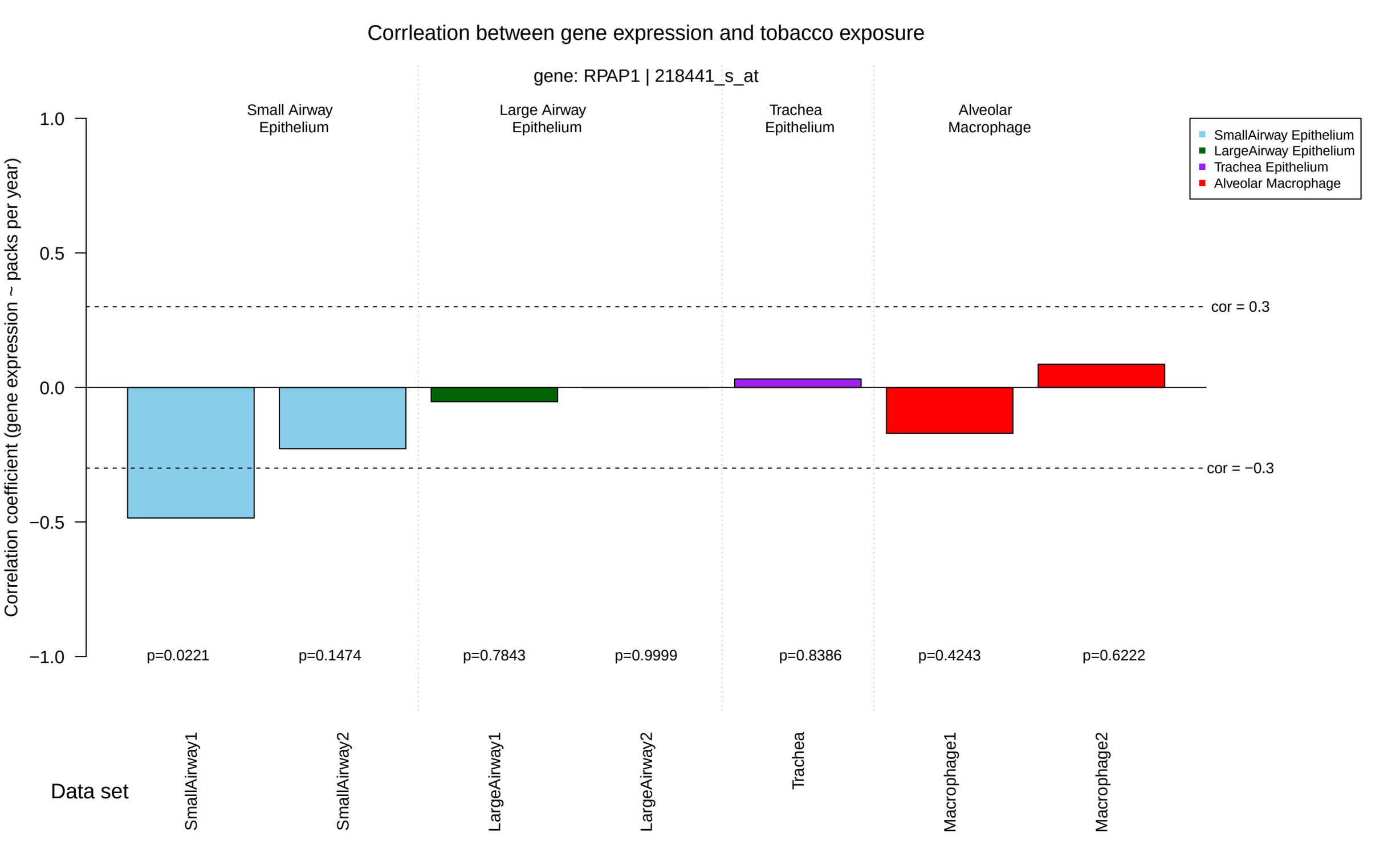

set: LargeAirway1 (p=0.6959)
RPAP1 | 218441_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5209)
RPAP1 | 218441_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4983)
RPAP1 | 218441_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4751)
RPAP1 | 218441_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1795)
RPAP1 | 218441_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPAP1 | 218441_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0221
p=0.1474
p=0.7843
p=0.9999
p=0.8386
p=0.4243
p=0.6222
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2